

MARTIN GUTHRIE

GIS Analyst | Fluvial Geomorphology & River Systems

martinguku37@gmail.com | +1 (406) 380 - 3324 | Columbia Falls, MT

SUMMARY

GIS Analyst with 5+ years of experience applying geospatial analysis, remote sensing, and terrain modelling to river systems, sediment dynamics, and the recovery of fluvial landscapes from disturbance. Background spans geoscience and mining contexts, with a particular interest in how human activity — including historical mining — reshapes river channels, floodplains, and sediment regimes over time.

Skilled across the full geospatial workflow using ArcGIS Pro, ArcMap, QGIS, ENVI, Python (ArcPy, GeoPandas), SQL, and PostGIS, with deep experience in LiDAR and DEM terrain analysis, channel-morphology extraction, geomorphic change detection, and catchment-scale spatial analysis. Comfortable building reproducible Python workflows that quantify channel change and sediment movement from reach to catchment scales.

Particular focus on sediment dynamics and mining-impacted rivers — mapping and modelling the mobilisation, storage, and re-deposition of fine and coarse sediment (including historical mine tailings and sand slugs) and characterising their long-term effects on channel form, riparian zones, and aquatic habitat. Experienced integrating drillhole, core, survey, geophysical, and imagery datasets into unified spatial and 3D models of valley-fill and sediment systems.

Collaborative and interdisciplinary by temperament, comfortable working at the boundary of physical and human geography alongside geomorphologists, ecologists, engineers, and archaeologists. Owns projects end-to-end — from data acquisition and QA/QC through analysis, 3D modelling, visualisation, and delivery — and translates complex spatial results into clear products for researchers, managers, and stakeholders.

Seeking to contribute geospatial and geomorphological analysis to river-science research, with a strong interest in river restoration, the recovery of disturbed fluvial systems, and the long-term legacy of mining on Australian and international rivers.

TECHNICAL SKILLS

Category	Skills & Tools
Fluvial Geomorphology & Hydrology	Catchment (drainage-basin) delineation, channel morphology & planform analysis, longitudinal & cross-sectional profiling, sediment budgets & transport pathways, flow accumulation & stream networks, floodplain & riparian mapping, geomorphic change detection
Remote Sensing & Change Detection	ENVI, ERDAS Imagine, multispectral & satellite imagery, historical aerial-photograph analysis, channel-migration & planform change detection, multitemporal / multidecadal analysis, land-cover & disturbance classification
Terrain & LiDAR Analysis	LiDAR point-cloud processing, DEM / DSM generation, DEM-of-Difference (DoD), hillshade & slope analysis, topographic & bathymetric change quantification, valley & floodplain characterisation
GIS & Mapping	ArcGIS Pro, ArcMap, QGIS, ArcGIS Online, spatial analysis, cartography & map production, georeferencing of historical maps, coordinate systems & projection management
Programming & Automation	Python (ArcPy, GeoPandas, NumPy, Pandas, rasterio), SQL, JavaScript, automation & batch scripting, ModelBuilder, reproducible geoprocessing workflows
3D & Subsurface Modelling	Leapfrog Geo, Petrel, SKUA-GOCAD, Move, valley-fill & sediment-body modelling, drillhole & core data integration, implicit surface & solid modelling
Spatial Databases & Storage	PostGIS, PostgreSQL, SQL Server, Oracle Spatial, spatial queries, geodatabase design, ETL & multi-source data integration

Category	Skills & Tools
Geostatistics & Data Science	Geostatistics, kriging, variography, sediment-property interpolation, statistical & exploratory spatial data analysis, machine-learning fundamentals
Web GIS & Visualisation	Mapbox, Leaflet, ArcGIS Online, interactive web maps & dashboards, thematic & 3D visualisation

EDUCATION

Bachelor of Science in Geographic Information Systems Technology

December 2020

University of Arizona • Tucson, AZ, US

PROFESSIONAL EXPERIENCE

NV5 Geospatial (Lakewood, CO, US)

GIS Analyst – Fluvial & Terrain Analysis

Aug 2023 – Mar 2026

Responsibilities

- Served as a primary GIS analyst delivering LiDAR-based terrain analysis and remote-sensing products for river-corridor, floodplain, and natural-resource projects across North America.
- Processed LiDAR point clouds and aerial imagery into high-accuracy DEMs, DSMs, and bathymetric surfaces used to characterise channel morphology, floodplain extent, and valley form.
- Quantified geomorphic change by generating DEM-of-Difference (DoD) surfaces to measure erosion, deposition, and sediment storage between survey epochs.
- Extracted channel centrelines, bank lines, longitudinal profiles, and cross-sections to support reach-scale geomorphic assessment.
- Mapped channel migration and planform change through multitemporal analysis of historical aerial photography and satellite imagery.
- Delineated catchments, stream networks, and flow-accumulation surfaces from DEMs to support hydrological and sediment-routing analysis.
- Automated repetitive terrain and change-detection workflows in Python (ArcPy, GeoPandas, rasterio), improving consistency and turnaround.
- Performed remote-sensing classification in ENVI to map riparian vegetation, land cover, and disturbance along river corridors.
- Designed PostGIS-backed spatial databases to manage large multitemporal terrain and imagery datasets with strict integrity requirements.
- Produced cartographic deliverables and figures communicating geomorphic and hydrological findings to technical and non-technical audiences.
- Conducted rigorous QA/QC on geometry, attributes, topology, and coordinate reference systems prior to analysis and delivery.
- Built interactive web maps and dashboards (ArcGIS Online, Mapbox, Leaflet) to visualise channel change and terrain analysis for stakeholders.
- Collaborated with engineers, surveyors, and environmental scientists to translate field and survey requirements into accurate geospatial products.
- Documented data lineage, methods, and metadata to ensure reproducibility and long-term maintainability.

Mira Geoscience (Montreal, QC, Canada)

GIS Analyst & 3D Modeller – Sediment & Subsurface

Apr 2022 – Jun 2023

Responsibilities

- Led GIS analysis and 3D modelling supporting mineral-exploration and earth-science projects, with a focus on characterising sediment bodies, valley fill, and the subsurface legacy of mining activity.
- Built 3D models in Leapfrog Geo, SKUA-GOCAD, and Move, generating implicit surfaces and solids representing sediment units, valley-fill stratigraphy, and lithological domains from drillhole and field data.
- Integrated drillhole, core, geochemical, geophysical, and survey datasets into unified spatial databases for multidisciplinary analysis.

- Applied geostatistical methods (variography, kriging) to interpolate sediment and material properties and delineate spatial domains.
- Characterised the distribution and volume of mining-derived materials and tailings, supporting assessment of their environmental and geomorphic footprint.
- Developed Python workflows to clean, transform, and validate large subsurface and sediment datasets, significantly reducing manual effort.
- Produced cross-sections, fence diagrams, and 3D visualisations communicating subsurface and sediment interpretations to geoscientists and clients.
- Designed geodatabase schemas and PostGIS storage for versioned exploration and sediment data with full traceability.
- Performed spatial analysis to map material distribution, assess prospectivity, and support resource and environmental evaluation.
- Conducted QA/QC on drillhole and survey databases, resolving collar, survey, and assay inconsistencies prior to modelling.
- Collaborated with geologists, engineers, and environmental scientists to translate physical concepts into reproducible digital models.
- Integrated remote-sensing and terrain datasets into broader catchment- and valley-scale interpretations.
- Standardised modelling workflows and documentation to support reproducibility and knowledge transfer across projects.
- Delivered milestones under tight timelines while maintaining rigorous accuracy and data-integrity standards.

SRK Consulting (Vancouver, BC, Canada)

GIS Analyst – Mining & Environmental

Feb 2021 – Jan 2022

Responsibilities

- Joined as a GIS Analyst after completing a B.Sc. in GIS Technology, supporting mining, environmental, and water-related consulting projects across a North American practice.
- Provided GIS analysis and cartographic support assessing the impacts of mining and land-use change on catchments, waterways, and floodplains.
- Conducted catchment (drainage-basin) analysis, preparing and interpreting terrain, hydrological, and geological datasets to characterise sediment sources and pathways.
- Digitised and georeferenced historical maps and aerial photographs to reconstruct past channel positions, mining disturbance, and landscape change over time.
- Mapped sediment, tailings, and disturbed ground to support environmental assessment of mining-affected river systems.
- Prepared thematic maps, figures, and spatial datasets for technical and regulatory reporting.
- Wrote introductory Python scripts to automate batch map exports, attribute updates, and repetitive geoprocessing tasks.
- Built and maintained spatial databases with consistent naming, metadata, and projection conventions.
- Integrated GPS and field-survey data into project GIS, ensuring positional accuracy and attribute completeness.
- Executed spatial queries and overlay analyses for site characterisation, environmental review, and catchment planning.
- Assisted with remote-sensing tasks, including imagery preparation and basic classification for land-cover and disturbance mapping.
- Conducted QA/QC on spatial data and map products before client and regulatory submission.
- Collaborated within multidisciplinary teams, aligning GIS outputs with technical and environmental objectives.
- Maintained organised, well-documented project data to support reproducibility and long-term maintainability.